# Artificial Intelligence Techniques *in* Manufacturing

# LLM

# Natural Language Processing (NLP)

- Branch of artificial intelligence (AI)
- Enables computers to comprehend, generate, and manipulate *human language*

# NLP Technologies

- Language modelling
  - Machine translation
  - Question answering
  - Text summarization
  - Text generation
- Speech
  - Text to speech
- Grounding
  - Text to image

Large Language Model (LLM)

# Text Generation

- If we want to model an image classifier
  - The task is "*predict the right label*".
- What should be the ***task*** for text generation?

# Next Word Prediction

The sky is → Probability distribution of all the next possible words

*blue 7%*
*the 6%*
*...*
*...*
*red 0.01%*

# Next Word Prediction

The sky is → **Language Model** → **Probability distribution of all the next possible words**

*Model that computes the probabilities.*

*blue 7%*
*the 6%*
*...*
*...*
*red 0.01%*

# N-Grams

# N-Grams

*How many times "the sky is blue" appears.*

$$P[blue \mid the\ sky\ is \mid \approx \frac{C[the\ sky\ is\ blue]}{C[the\ sky\ is]}$$

*How many times "the sky is" appears.*

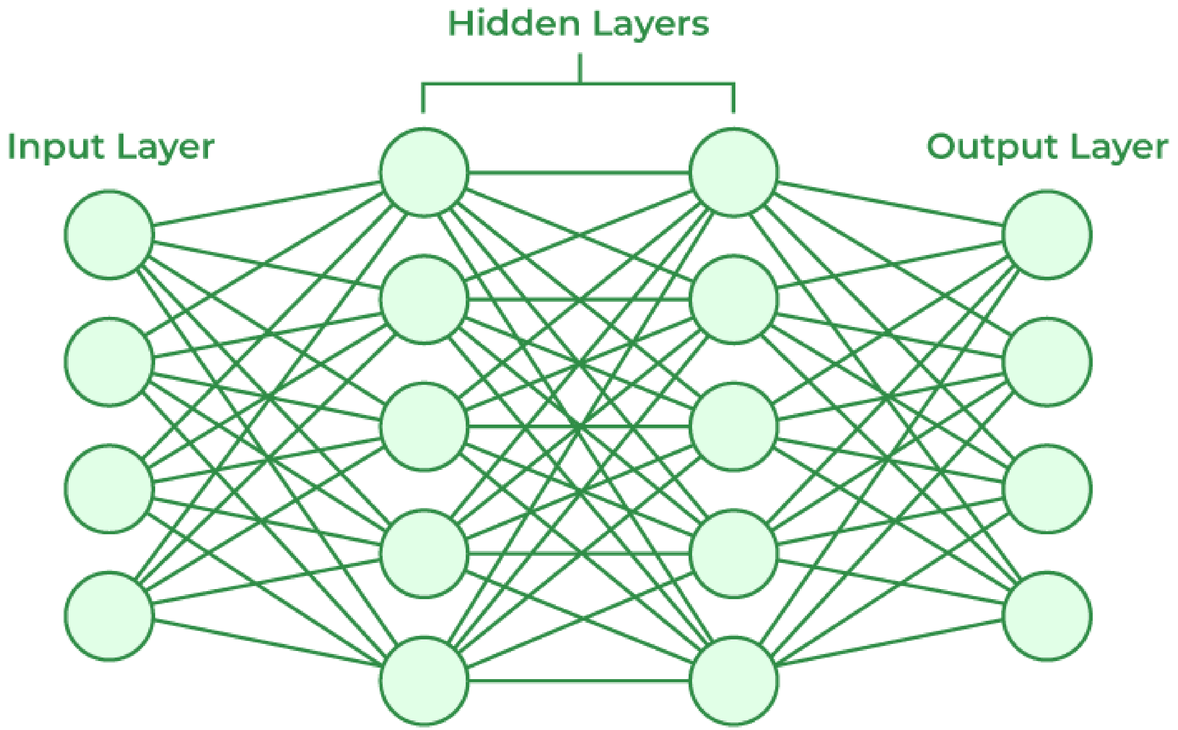
# N-Grams Disadvantages

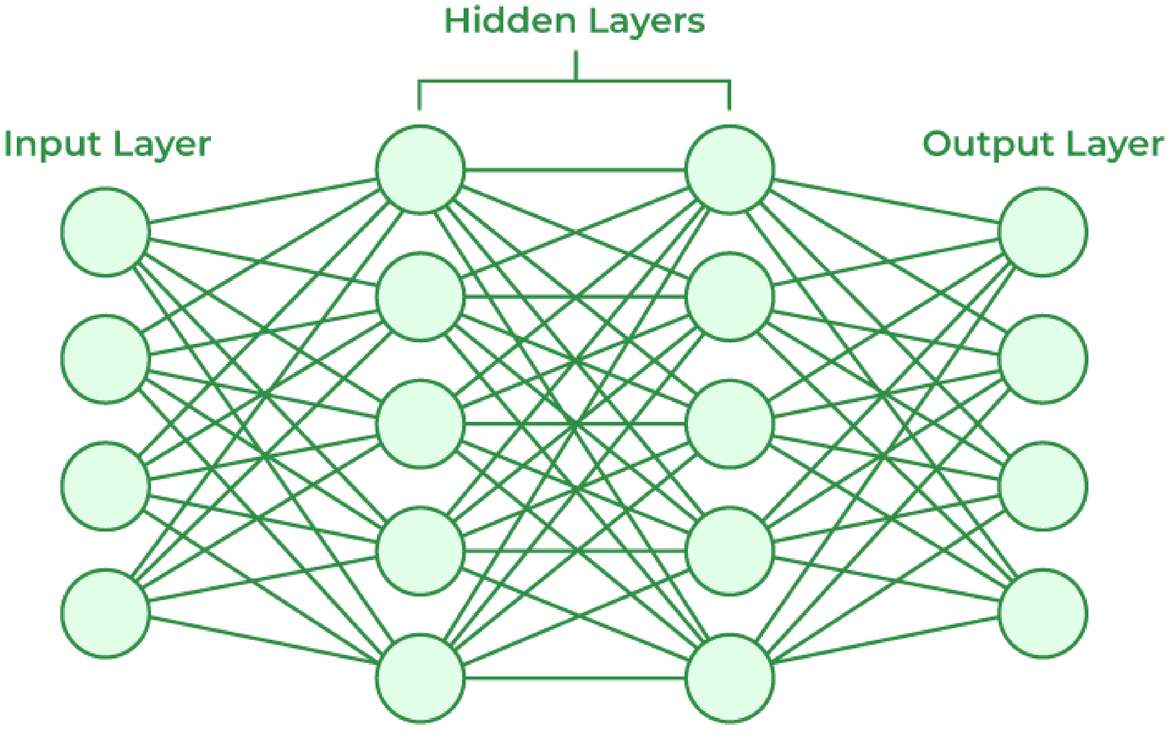
- Too specific
  - What if we have “the sky was”
- Context lost
  - Cannot “remember” words that are far away

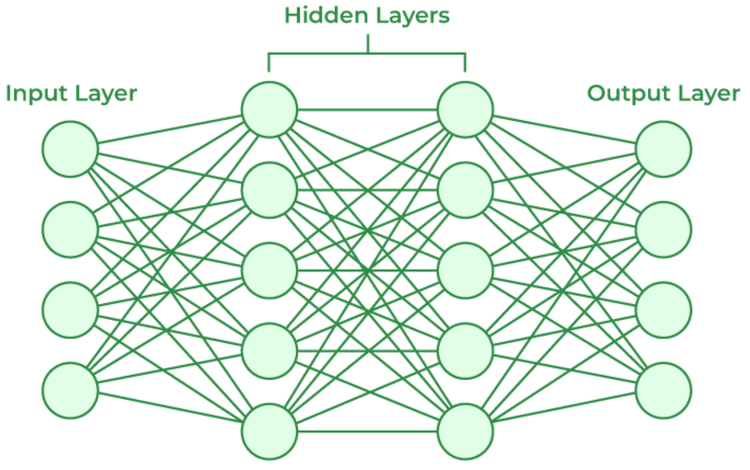
# Recurrent Neural Network

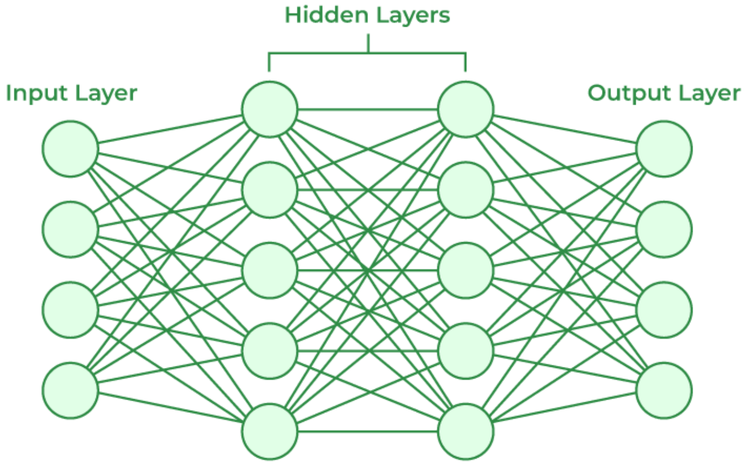
# RNN

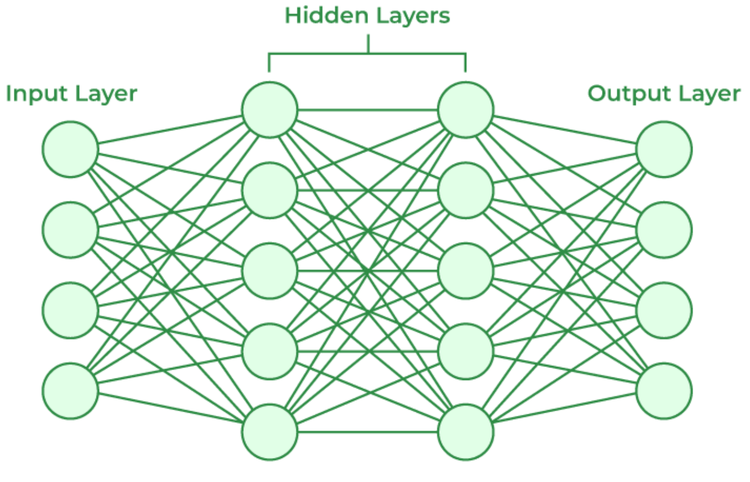
- A class of neural networks that can process sequences of data
  - Maintains hidden states that capture information about previous time steps.
- RNN can maintain memory of previous inputs.

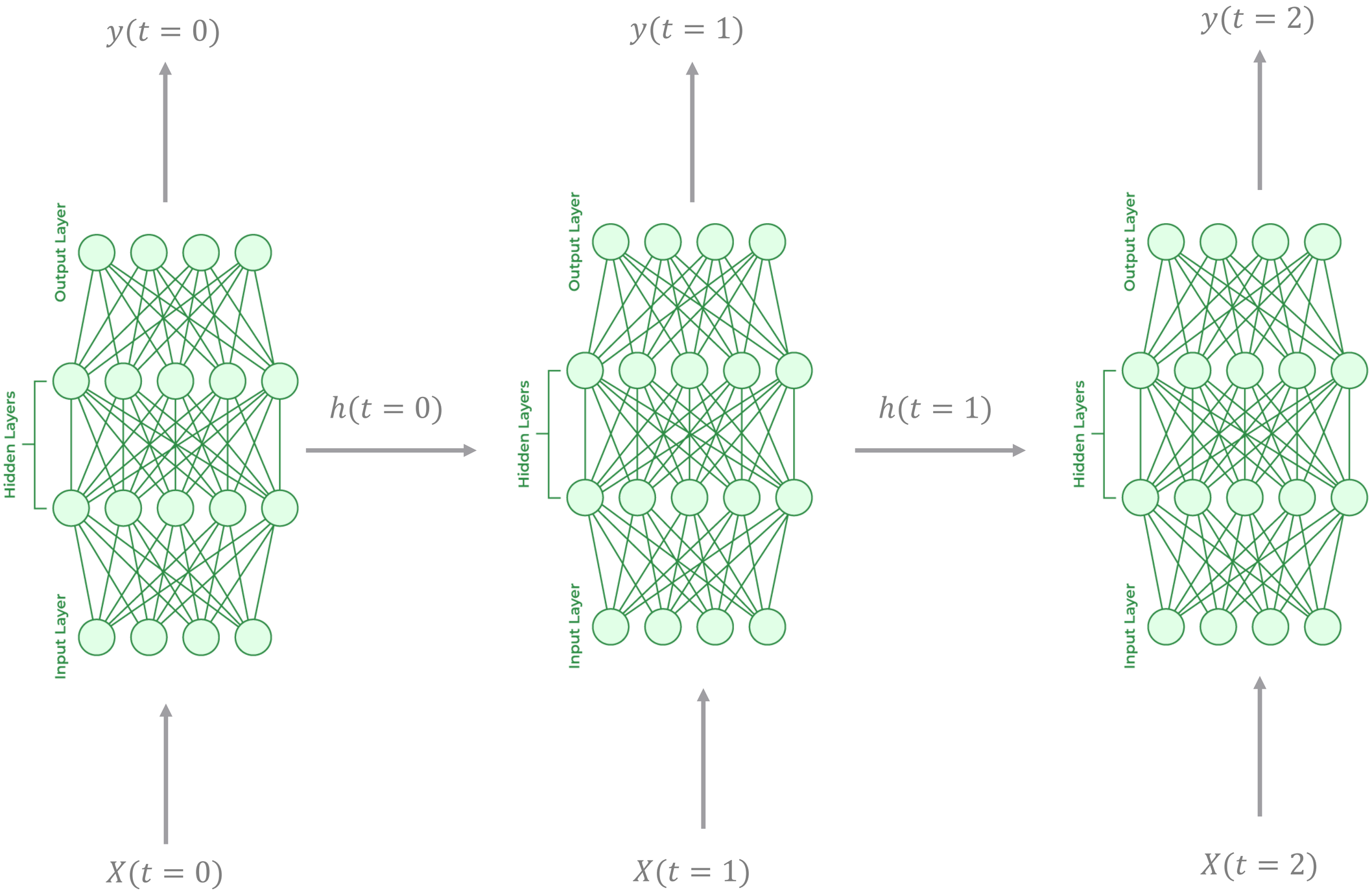

Hidden Layers
Input Layer
Output Layer


Hidden Layers
Input Layer
Output Layer


Hidden Layers
Input Layer
Output Layer


Hidden Layers
Input Layer
Output Layer

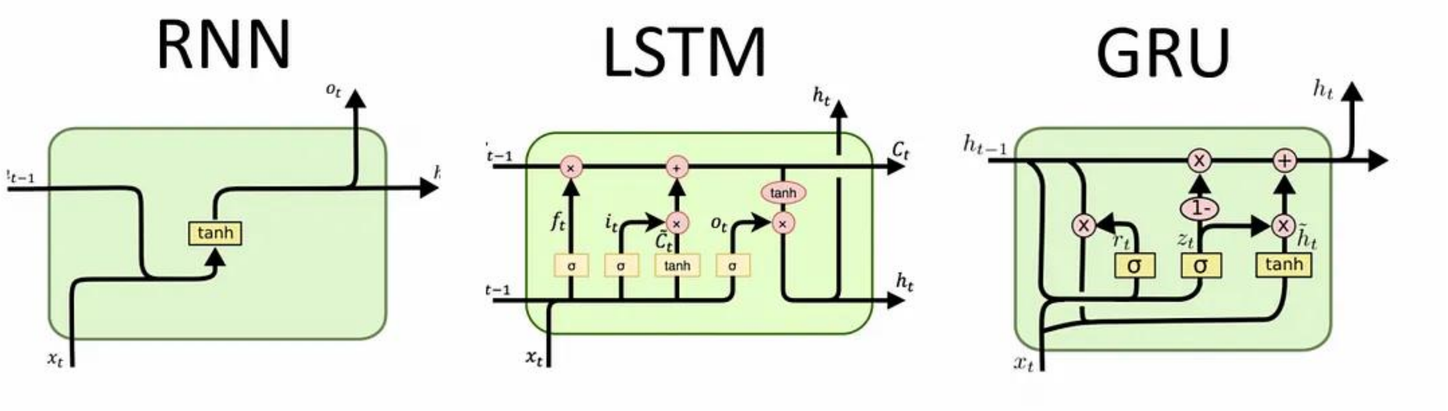

Hidden Layers
Input Layer
Output Layer


$y(t=0)$
$y(t=1)$
$y(t=2)$
Output Layer
Hidden Layers
Input Layer
$h(t=0)$
$h(t=1)$
$X(t=0)$
$X(t=1)$
$X(t=2)$

# Resources

- Diagram
  - https://colah.github.io/posts/2015-08-Understanding-LSTMs/
- Animation
  - https://joshvarty.github.io/VisualizingRNNs/
- Good Read
  - https://karpathy.github.io/2015/05/21/rnn-effectiveness

# RNN Variants




https://medium.com/@hasanshujaat4/recurrent-neural-network-and-its-variants-de75f9ee063

# RNN Disadvantages

- Context lost
  - Vanishing gradient
- Sequential calculation
  - Cannot parallelize computation

# Transformer

# Transformer

- A type of neural network architecture that excels at processing sequential data
  - Utilizing a mechanism called ***self-attention*** to understand the context and relationships between different parts of the sequence.

# Transformer Models

- Encoder-Decoder
  - Language translation
- Encoder (only)
  - Bert (Bidirectional Encoder Representations from Transformers)
- Decoder (only)
  - GPT (Generative Pretrained Transformer)

# Resources

- Visualization 1
  - https://poloclub.github.io/transformer-explainer/
- Visualization 2
  - https://bbycroft.net/llm
- Attention Visualized
  - https://huggingface.co/spaces/exbert-project/exbert
- Really good lectures
  - https://www.nlpdemystified.org/course